

Geometry 6.5 — Homework Answers

Odd-numbered exercises. Check your work against these; bring questions on the evens to class.

1. the largest angle is opposite the longest side: $\angle A$ (opposite 7) $<$ $\angle B$ (opposite 9) $<$ $\angle C$ (opposite 11)
3. $14 > 13$ ✓ (the other two sums are clearly large enough) — YES
5. 16, which is not GREATER than 16 — NO (the three points would be collinear)
7. $12 - 12 < x < 12 + 12$, so $0 < x < 24$
9. Each obtuse angle measures more than 90° , so two of them total more than 180° . Adding the positive third angle makes the sum greater than 180° , contradicting the Triangle Sum Theorem. The assumption must be false, so a triangle has at most one obtuse angle.